

STAT 654

Chapter 1: Introduction to Statistical Learning

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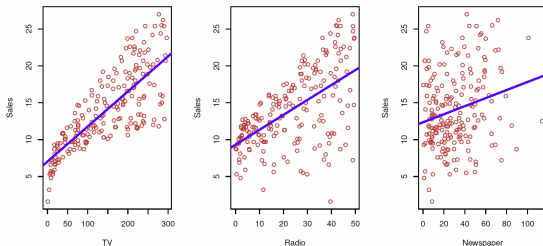
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An Overview of Statistical Learning

- *Statistical Learning* refers to a vast set of tools to understand data.
- These tools can be classified as *supervised* or *unsupervised*.
- *Supervised* statistical learning involves models for prediction or estimation.
- Alternatively, an *output* based on *input(s)* (e.g., regression) .
- *Unsupervised* statistical learning involves inputs but no outputs (e.g., a clustering problem).

Example of Statistical Learning: Advertising and Sales

- Goal: Investigate association between advertising and sales of a product (Refer to [Advertising](#) data of ISLR2).
- Information on sales in 200 markets, and ad budgets in TV, radio and newspaper.

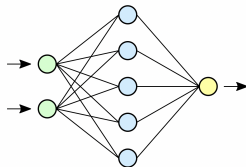


Sales vs. Ad costs in 3 mediums. The blue line in each figure represents the simple least squares line.

- Response Y and p predictors, $(X_1, X_2, \dots, X_p)$
- Assume there is some relationship between Y and $\mathbf{X}$, with general form $Y = f(\mathbf{X}) + \epsilon$.
- Here f is a fixed but unknown function of $(X_1, X_2, \dots, X_p)$.
- Represents the *systematic* information $\mathbf{X}$ provides about Y .
- Can be used for **prediction**.
- ϵ is a random *error term*. Independent of $\mathbf{X}$ and mean 0.

Prediction

- Estimating f is necessary for **prediction** and **inference**.
- **Prediction:** $\mathbf{X}$ available, but no corresponding Y value for some observations.
- Since error term averages to 0, predict Y using $\hat{Y} = \hat{f}(\mathbf{X})$.
- In this setting, $\hat{f}$ may be treated as a *black box*, i.e., we are not typically concerned about the exact form of $\hat{f}$.
- Not drawing any inference on $\hat{f}$, i.e., not interested in estimate, CI of $\hat{f}$ (for e.g., a neural network). Only prediction.



A Neural Network (Black Box)

Inference

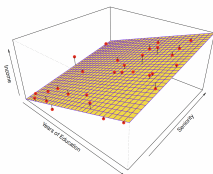
- Estimating f is necessary for **prediction** and **inference**.
- **Inference**: Understand the association between Y and $\mathbf{X}$.
- We wish to estimate f . Goal not necessarily to make predictions. Cannot treat $\hat{f}$ as a black box.
- The following important questions are relevant in this context:
 - Which predictors are associated with the response? (i.e., identify the *important* predictors.
 - What is the relationship between response and predictor(s) (i.e., +ve, -ve, interaction effects)?
 - What is the correct form of f ?

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Parametric Methods

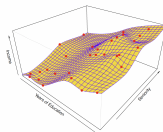
- Parametric methods involve a 2-step model-based approach:
 - ① Make assumption about the functional form of f . A simple assumption is that f is linear in $\mathbf{X}$ (**Linear model**), i.e.,
$$f(\mathbf{X}) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$$
 - ② Use *training* data to *fit* or *train* the model. Estimate the parameters $\beta_0, \beta_1, \dots, \beta_p$. (e.g., use ordinary least squares).
- **Pro:** Simpler to estimate f . **Con:** Fitted model may be very different from true f .



A linear model (with no interactions) fit by least squares to Income data (ISLR). Observations are in red. Yellow plane is the least squares fit.

Non-Parametric Methods

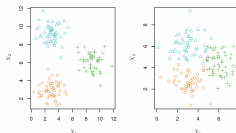
- Non-parametric methods do not make explicit assumptions about the functional form of f .
- Seek an estimate of f that gets as close to the data points as possible without being too rough.
- Advantage: No assumption on functional form of $f \Rightarrow$ Potential to accurately fit a wider range of possible shapes for f .
- Disadvantage: Do not reduce the problem of estimating f to a small number of parameters $\Rightarrow$ Large number of observations (far more than is typically needed for a parametric approach) required in order to obtain an accurate estimate for f .
- Examples of non-parametric methods: Splines, Gaussian process, Dirichlet process.



A spline model fit to Income data (ISLR)

Supervised Vs. Unsupervised Learning

- **Supervised learning:** examples that we have discussed so far. For predictor measurement(s) x_i , $i = 1, \dots, n$ there is an associated y_i .
- Model relating response to predictors for prediction or inference.
- **Unsupervised learning:** For every observation $i = 1, \dots, n$, we observe a vector of measurements x_i but no associated response y_i .
- Cannot fit a linear regression model, since there is no response variable to predict.
- Goal: Understand relationships between the variables or between the observations.
- E.g., Clustering: ascertain, on the basis of $x_1, \dots, x_n$, whether the observations fall into analysis relatively distinct groups.



Clustering data sets involving three groups. In practice, group memberships may be unknown.

Assessing Model Accuracy

- No one method dominates all others over all possible data sets.
- Hence it is an important task to decide for any given set of data which method produces the best results.
- Selecting the best approach can be one of the most challenging parts of performing statistical learning in practice.
- Need to measure how well a method's predictions actually match the observed data.
- In a regression setting, most commonly-used measure is the *Mean Squared Error* (MSE).
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$
, where $\hat{f}(x_i)$ is the prediction.
- Can use MSE for *training* as well as **test** data (more important - e.g., stock data, prediction of disease risk).
- We want to choose the method that gives the lowest test MSE.

The Classification Setting

- Now $y_1, \dots, y_n$ are qualitative, and we want to estimate f .
- To quantify the accuracy of $\hat{f}$, we use the training *error rate*, the proportion of mistakes that are made if we apply our estimate $\hat{f}$ to the training observations:
 $\frac{1}{n} \sum_{i=1}^n I(y_i \neq \hat{y}_i)$ ($\Rightarrow$ the fraction of incorrect classifications).
- If $I(y_i \neq \hat{y}_i) = 0$ then the i -th observation was classified correctly by our variable classification method; otherwise it was misclassified.
- Similar idea for the *test error rate* for test data.
- The test error rate for a set of test observations of the form (x_0, y_0) is $\text{Ave}(I(y_0 \neq \hat{y}_0))$, where $\hat{y}_0$ is the predicted class label.
- Would like to minimize the test error to get a *good* classifier.

An Introduction to Statistical Learning by G.James, D.Witten,
T.Hastie, R.Tibshirani